# *SciRoute*: From Scientific Queries to Native File Access

Supplementary Experimental Results

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## S1 EXPERIMENTAL METHODOLOGY

This supplement provides additional protocols and results for the main paper's Section 6, Experimental Evaluation. Each section identifies the main-paper study it expands. This section details the resources, timing boundaries, repetition schedules and statistical definitions underlying Section 6.1. Source code is available at https://github.com/thulab/SciRoute.

*Comparable Work.* FS-scan and FS-index share numerical readers; FS-index reuses discovery. IoTDB-FS integrates publication, retention, validation, format execution and typed delivery. The single-file campaign separately compares Native FS with managed modes; its baseline is not automatically FS-scan. Reader state is matched in the discovery controls; comparisons with retained handles or decoded state measure the complete configured paths. Historical FS-scan and the HDF5 handle-reuse follow-up were collected later on matched inputs, as detailed in Sections S6 and S7. External mapping versions remain fixed throughout each campaign.

*Resource Configuration.* All campaigns run on a Kernel-based Virtual Machine (KVM) with 32 virtual processors. Original local GRIB/HDF5 runs record approximately 378 GiB of RAM; the stored-data remeasurement records 189 GiB. Resource limits are matched within campaigns:

- Original local GRIB/HDF5 queries share two cores among the client, service, native helper, and DataNode. Query memory is limited to 2 GiB; DataNode heap/direct memory is separately limited to 15,232/2,176 MiB.
- Arrow, stored-data remeasurement, and incremental registration retain the shared two-core and 2 GiB query budget, with DataNode heap/direct limits of 28/4 GiB.
- Offline index construction uses two exclusive logical cores and 2 GiB. Remote-access query/storage containers occupy the same host with separate two-core budgets and 2/24 GiB memory limits; Section S8 specifies the network controls.

The Arrow campaign records Python 3.12.3, PyArrow 21.0.0, NumPy 2.1.3, GCC 13.3.0 and ext4 storage. The HDF5 handle-reuse follow-up uses HDF5 1.14.4. Local query swap is disabled; ConfigNode remains outside the query quota. Operating-system and database caches remain enabled. Each study identifies reader reopening or reuse; initial requests retain operating-system caches. Measurements were collected during normal high-load operation with ongoing model training. Reported dispersion describes observed system variability.

*Measurement Boundaries.* Query time spans discovery through client-owned values and validity masks. All paths select sources before numerical reading. With discovery $d$ and reading, decoding and delivery $q$, each request satisfies $T = d + q$; independently summarized phases need not sum to the total summary. Preparation, publication and startup precede warm-query clocks. Single-file independent sessions include preparation, fresh requests and cleanup. Stored-data sessions measure serving over query-ready sources and prepared indexes; source verification, recovery and auditing lie outside their serving clocks. Publication-to-result lifecycles and complete incremental traces charge their stated publication and finalization work, with boundaries in Section S7. Numerical and encoded reference checks cover source identity, values, masks, shapes, coordinates, ordering and digests as applicable to each study.

*Workload Notation.* In collections, $N$ counts total files, $C$ candidate logical paths, $K$ returned native units, and $S$ the original-file size profile, including nonmatching payloads. Histories vary $V$, retained versions per logical path. The notation $Q$ is study-specific: it denotes in-flight requests in collection/remote concurrency comparisons, requests per session, or total queries per incremental trace ($Q = 16q$, with $q$ queries per update, distinct from the reading phase above). Remote tests vary added round-trip delay $L$. Values and schedules accompany each study. The remote table writes $c$ for the in-flight count; collection tables label the one- and four-request schedules directly.

Table S1 retains the recorded summaries and complements the main paper's study-specific statistical definitions. Table S4 distinguishes construction means and medians.

*Main-Paper Dispersion.* Collection plots report request medians and observed minimum–maximum ranges over ten measured requests for each native path and forty requests from ten sequentially submitted four-query batches for the managed path. The range endpoints depend on these respective sample counts. Historical cells summarize three round medians by their median and sample standard deviation (with denominator $3 - 1$). The FS-scan history measurements come from a later matched-input collection. These

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**Table S1: Measurement repetitions and summary statistics for each experiment.**

| Study | Measurement schedule | Reported summary |
|---|---|---|
| Construction | Three fresh-process builds per input | Arithmetic mean, median and range. |
| Single-file | Three rounds; two warmups and ten measured queries per round | Median of round medians; complete sessions and lifecycles reported separately. |
| Arrow | Three warmups and thirty rotating measurements per path/configuration | Query median; controls retain their own reader state. |
| Collection | One campaign; two warmups, then ten native queries or ten four-query managed batches | Request median; batch throughput from completed work. |
| History | Three rounds; two warmups and ten queries per case | Median of round medians. |
| Remote | Initial batch, two warmup batches and ten measured four-query batches | Query median and measured throughput. |
| Stored-data sessions | Three rounds per source/path/query count | Arithmetic mean of session wall time divided by query count. |
| Incremental maintenance | Three complete traces per format, query count and path | Arithmetic mean of trace totals. |

ranges and standard deviations are descriptive statistics of within-campaign variability at each study's sampling level.

## S2 ACCESS PATH CONSTRUCTION

This section expands Section 6.2.1 of the main paper with semantic mapping checks and index construction measurements.

Semantic mappings make discovery useful when a named quantity and its coordinates identify the intended native data. Table S2 summarizes source-level checks alongside numerical agreement. Source records establish meaning; native references check the returned values and masks. These complementary checks support the timing comparisons on matched inputs.

Temperature–Rain keeps its source and derived time labels distinct and uses ordinal windows. Appliances retains civil timestamps without assigning an unverified time zone. LOOP, Solar and KDD use publisher identifiers and ordinal coordinates where scientific meanings remain unresolved. GEOS uses physical floating-point values under its source-specific range interpretation. Selections use only quantities and coordinates supported by the source evidence.

*Functional Validation.* Beijing/Jena evaluation checks six development queries and twelve later instances on all three paths, giving 54 passing path/query checks. A separate boundary check passes all nine combinations of three paths and three cases: a Jena gap, a duplicate timestamp, and a Beijing batch boundary. Table S3 adds the archived functional evidence for publication selection, residual inputs, and source admission. Its catalog, module, and real-backend checks establish these behaviors at their respective layers.

**Table S2: Source and value checks used to validate scientific mappings. Counts describe checked observations and source coverage.**

| Input | Semantic mapping evidence and native access |
|---|---|
| GRIB | Quantity, unit, level and forecast times on sources with 22 and 4,598 messages; encoded locations and native numerical values. |
| Sea-surface temperature | Seven daily files and 56 catalog entries; seven positive and zero absent-label matches; 1,680 value/mask observations across five access paths. |
| GEOS atmospheric data | Two time slices, four documented coordinate interpretations and twelve point/region checks; native origins, shapes and physical values agree. |
| Arrow sources | All 23,741 Saugeen values; all 19,735 Appliances rows and 29 columns; 23,252,200 Temperature–Rain value/missing positions. |

**Table S3: Checks of publication selection, query evaluation, and source access. All listed checks pass within the validation scopes shown.**

| Input condition | Expected and observed behavior | Validation scope |
|---|---|---|
| Old version matches; newest does not | Latest selects newest first, then returns no match | In-memory catalog test |
| Residual field omitted from projection | Read hidden values/masks, filter, then remove hidden fields | Arrow module, in-memory backend |
| New publication after query preparation | Prepared context retains its original exact source | HDF5 real-backend context checks |
| Revoke a pruned candidate's or ancestor's permissions; delete a frozen source | Reject new admission even if that source contributes no output | HDF5 backend, non-root checks |

*Agent-Assisted Construction.* The Beijing and Jena artifacts were developed in one Codex-assisted session. Data, documentation, existing native readers and the environment were prepared beforehand. The recorded interval from formal construction start to freezing semantic mappings, indexes and reader adaptations is 986.472 s (16.4 min), including repairs within that interval. Earlier exploratory work preceded this interval, and reference evaluation and formal query measurements followed the freeze. Twelve query instances were withheld until the freeze and evaluated afterward within the same session. The same Codex workflow performed development and evaluation, using separate CSV/TSF parsing logic for source-record comparisons.

*Index Rebuilding Cost.* The construction clock measures complete accepted-package rebuilding, including source hashing, semantic/layout checks, serialization, and file/directory synchronization. Semantic-mapping checks are shared preparation for native and indexed access. Acquisition, agent development, process imports, independent reference validation and managed publication lie outside this clock. Three fresh-process builds per input use two processor cores and 2 GiB memory, with retained operating-system caches. Beijing checks 420,768 calendar rows; Jena checks 1,106,616 source values ($21 \times 52,696$ records) and their timestamp and missing-value alignment. Jena's records include one identical duplicate; 52,695 unique times and nine gaps form the 52,704-position grid. After timing, each rebuilt index passes the eighteen query instances and three boundary cases on Native FS and FS-index: $3 \times 21 \times 2 = 126$ checks. Reconstruction checks cover Native FS and FS-index. IoTDB-FS results come from the earlier three-path evaluation.

**Table S4: Index rebuilding time and package size over three builds. Times are in seconds, rounded to three decimals. Package size counts serialized bytes.**

| Input | Mean | Median | Range | Package bytes |
|---|---|---|---|---|
| Beijing | 1.453 | 1.446 | 1.422–1.489 | 38,568 |
| Jena | 4.228 | 4.308 | 3.975–4.401 | 11,208 |

The packages occupy 0.072% and 0.253% of their Arrow source sizes. Table S4 reports median rebuild times of 1.446 s for Beijing and 4.308 s for Jena, as used in the main paper.

## S3 ARROW QUERY EVALUATION

This section provides the complete Arrow configurations and access controls for main-paper Sections 6.2.2 and 6.3.1.

The Arrow study asks how much locating a scientific object saves when its values remain in native storage. Four query classes across six inputs and two native-reader states yield 48 configurations in one campaign. Each path receives three warmups and thirty rotating measurements per configuration, with balanced path order, totaling 4,320 measured results. All outputs match an independently materialized reference. Query clocks include complete values and masks; conversion, publication, index construction and reference generation precede them.

*Query Workloads.* The five nested inputs use the published GIFT-Eval snapshot. Each outer row pairs `item_id` with a Float32 list in `target`. The flat UCI Appliances conversion retains CSV order, timestamp-second coordinates and Float64 measurements, with 4,096 rows per full batch. Table S5 records the layouts and selected windows; all six inputs have uncompressed buffers. Thus nested outer-row counts and time-series lengths describe different dimensions of the workload.

ID/time selects an object and a 128-step window; Position supplies the same window's native location directly. Full returns the complete selected series, or all rows of the projected Appliances columns. For nested inputs, the projection is `target` and Filter retains valid values greater than zero over the selected series' full range. Appliances projects `Appliances` (energy, Wh) and `T1` (temperature, degrees Celsius). Its Filter query retains rows whose valid energy values exceed 100 Wh. These thresholds exercise residual evaluation rather than domain-specific event definitions. The same client-side filter is applied to owned values and masks on all three paths; NaNs do not satisfy the ordered comparisons. Results include ordinal positions and exact source identity.

For Appliances, ID/time uses the civil-time interval [2016-03-20 05:30, 2016-03-21 02:50), corresponding to the ordinal window in Table S5; no time zone is inferred. Full and Filter cover [2016-01-11 17:00, 2016-05-27 18:10). The two native paths share their numerical reader and retain the same batches within each condition. IoTDB-FS freshly registers and owns results in both conditions. Reader state below therefore refers to the native contexts, while managed execution retains its own request contract.

With reopened readers, the known-position control still includes reader initialization and result delivery. Its native/index ratios range from 1.12 to 2.42; ID/time ratios range from 1.39 to 6.16. These ratios describe the complete execution paths under the reopened-reader condition. Tables S6 and S7 retain all query classes and reader conditions, including comparisons close to parity.

### S3.1 Query Costs with Retained Readers

With retained readers, native/index ratios for Position range from 0.91 to 1.11. ID/time ratios range from 1.40 to 8.48; the Temperature–Rain and Appliances ID/time windows reach 2.97 and 8.48. Within-class ratios show near-parity for known-position access and larger gains for ID/time selection. Absolute latency can be higher for Position than for ID/time; subtracting these separately measured classes would not isolate a locating cost. IoTDB-FS has higher median latency than Native FS in all 48 configurations: managed validation and delivery remain part of complete query cost.

The independent whole-file count control compares the two native paths; the managed nested interface selects one row per request and has no implemented whole-file count adapter. The control retains every candidate value across all six inputs. With retained readers, its native/index ratios range from 0.974 to 1.024; Temperature–Rain inspects all 23,252,200 candidate values in each path. These results locate the benefit in discovery and positioning while leaving value-dependent scanning work. This two-path control has its own reader/result packaging and timing envelope; its latencies are kept separate from the three-path study.

*Additional Arrow Workloads.* The separate construction study in Section S2 uses two further layouts. Beijing uses flat station files with nine batches and 35,064 hourly rows each; Jena stores one nested row with 21 channels of 52,704 ten-minute positions. For example, the Beijing threshold query projects PM2.5 and temperature for Aotizhongxin over January 2015, retaining PM2.5$>$ 75 $\mu$g/m$^3$ and temperature$<$ 0 degrees Celsius. The Jena threshold query selects temperature and relative humidity over ordinal positions [28800, 29808), retaining temperature$>$ 25 degrees Celsius and humidity$<$ 60%. The other templates select one station/window or multiple stations in Beijing, and one or two ordinal windows in Jena. Each of these six templates has one development instance and two later parameter instances. Across these eighteen cases, managed latency spans 2.64–85.05 ms and exceeds native latency

**Table S5: Arrow file layouts and query windows for both native-reader states. MB denotes $10^6$ bytes. Batch and row indexes start at zero, and $n$ is the series or table length. ID/time and Position use the same listed window. Position supplies batch/row locations for nested data or global row bounds for Appliances. Full and Filter cover $[0, n)$.**

| Dataset | MB | Outer rows | Batches | Selected ID | Batch/row | $n$ | Window |
|---|---|---|---|---|---|---|---|
| LOOP | 11.327 | 323 | 1 | `169` | 0/161 | 8,760 | [4380,4508) |
| Solar | 28.808 | 137 | 1 | `item_68` | 0/68 | 52,560 | [26280,26408) |
| KDD | 11.777 | 270 | 1 | `T136` | 0/135 | 10,898 | [5449,5577) |
| Saugeen | 0.096 | 1 | 1 | `T1` | 0/0 | 23,741 | [11870,11998) |
| Temp.–Rain | 93.877 | 32,072 | 33 | `T16036` | 16/36 | 725 | [362,490) |
| Appliances | 4.587 | 19,735 | 5 | — | — | 19,735 | [9867,9995) |

**Table S6: Arrow query latency when native readers are reopened for each request (24 configurations). Entries are medians over 30 requests, in ms, rounded to three decimals. Bold marks the lowest unrounded median per row.**

| Dataset | Query | Native FS | FS-index | IoTDB-FS |
|---|---|---|---|---|
| LOOP | ID/time | 0.640 | **0.421** | 2.627 |
| | Position | 0.660 | **0.546** | 3.057 |
| | Full | 0.663 | **0.468** | 3.718 |
| | Filter | 0.854 | **0.666** | 4.073 |
| Solar | ID/time | 0.611 | **0.421** | 2.884 |
| | Position | 0.658 | **0.587** | 3.170 |
| | Full | 1.176 | **0.922** | 8.831 |
| | Filter | 1.049 | **0.840** | 7.722 |
| KDD | ID/time | 0.863 | **0.622** | 3.177 |
| | Position | 0.345 | **0.295** | 2.219 |
| | Full | 0.948 | **0.729** | 4.612 |
| | Filter | 0.844 | **0.637** | 4.277 |
| Saugeen | ID/time | 0.787 | **0.546** | 3.154 |
| | Position | 0.357 | **0.284** | 2.297 |
| | Full | 1.248 | **1.004** | 7.171 |
| | Filter | 0.795 | **0.585** | 5.287 |
| Temp.–Rain | ID/time | 2.146 | **0.656** | 8.415 |
| | Position | 1.393 | **0.576** | 7.710 |
| | Full | 2.270 | **0.636** | 11.451 |
| | Filter | 2.088 | **0.522** | 8.247 |
| Appliances | ID/time | 3.806 | **0.618** | 7.651 |
| | Position | 0.631 | **0.350** | 5.683 |
| | Full | 4.754 | **1.848** | 40.113 |
| | Filter | 4.454 | **1.763** | 36.341 |

**Table S7: Arrow query latency when native readers are reused across requests (24 configurations). Entries are medians over 30 requests, in ms, rounded to three decimals. Bold marks the lowest unrounded median per row.**

| Dataset | Query | Native FS | FS-index | IoTDB-FS |
|---|---|---|---|---|
| LOOP | ID/time | 0.442 | **0.278** | 3.647 |
| | Position | 0.176 | **0.159** | 2.809 |
| | Full | 0.588 | **0.463** | 4.978 |
| | Filter | 0.369 | **0.254** | 4.225 |
| Solar | ID/time | 0.460 | **0.328** | 3.811 |
| | Position | **0.157** | 0.173 | 3.220 |
| | Full | 0.785 | **0.619** | 8.975 |
| | Filter | 0.956 | **0.839** | 9.744 |
| KDD | ID/time | 0.217 | **0.141** | 2.657 |
| | Position | **0.305** | 0.332 | 3.738 |
| | Full | 0.295 | **0.216** | 3.721 |
| | Filter | 0.609 | **0.480** | 5.119 |
| Saugeen | ID/time | 0.217 | **0.148** | 2.602 |
| | Position | **0.345** | 0.353 | 3.977 |
| | Full | 0.425 | **0.328** | 5.206 |
| | Filter | 0.679 | **0.571** | 6.292 |
| Temp.–Rain | ID/time | 0.654 | **0.220** | 8.187 |
| | Position | **0.176** | 0.185 | 7.185 |
| | Full | 0.858 | **0.355** | 9.039 |
| | Filter | 0.689 | **0.243** | 7.970 |
| Appliances | ID/time | 2.913 | **0.343** | 8.397 |
| | Position | 0.207 | **0.196** | 6.389 |
| | Full | 3.776 | **1.060** | 40.060 |
| | Filter | 3.730 | **1.020** | 35.629 |

in every case. These timings retain their own campaign and are not pooled with the six-input study.

## S4 SINGLE-FILE QUERY COSTS

This section provides the native-granularity evidence for main-paper Section 6.2.2 and extends the single-file selections and reader-reuse measurements in Section 6.3.2.

This study examines how native granularity, output size and reader reuse affect complete query cost. Table S8 expands the main paper's narrow selections with larger regions, encoded output, array sizes and projections. Native FS reads original files. The managed native mode uses the HDF5 library over complete managed files; managed layout uses the maintained format layout and payload. These modes form a separate study from external FS-index. The main paper's narrow selections use a 537.68 MB HDF5 file and one Electricity Arrow channel over 128 steps. Retained daemons share 256 MiB of metadata/decoded state, while ordinary Native FS reopens files with library defaults. Operating-system and database caches remain enabled. Each round has an initial request, two warmups and ten measured requests, with complete result consumption and independent result checks.

The HDF5 regions return 2,016 values from two chunks and 192 values from eight chunks: output size and accessed native units

**Table S8: Single-file query latency for additional selections (ms). Entries are medians of three round medians. A dash denotes an unevaluated mode, and $t$ counts HDF5 time slices.**

| Workload | Native FS | Managed native | Managed layout |
|---|---|---|---|
| GRIB 22 messages, point | 15.695 | — | 2.073 |
| GRIB 4,598 messages, region | 1426.278 | — | 4.404 |
| GRIB 4,598 messages, encoded | 1416.747 | — | 4.442 |
| HDF5 point, $t = 2$ | 0.397 | 2.858 | 2.167 |
| HDF5 point, $t = 1024$ | 0.434 | 3.230 | 2.400 |
| HDF5 region, 2,016 values | 0.461 | 3.059 | 2.282 |
| HDF5 region, 192 values | 0.772 | 3.360 | 2.273 |
| GEOS point | 1.232 | 4.139 | 2.952 |
| GEOS region, 256 values | 1.723 | 4.734 | 3.008 |
| OISST region, 64 values | 7.552 | 10.415 | 2.778 |
| Arrow 370 × 128 | 11.935 | — | 31.408 |
| Arrow 370 × 4096 | 16.780 | — | 91.494 |
| Arrow 1 × 128, uncompressed | 0.339 | — | 2.694 |
| Arrow 1 × 128, compressed | 14.950 | — | 2.714 |

differ. GEOS point/region requests touch one/four of 2,688 chunks, while both OISST selections touch the same chunk; managed layout takes 3.653/2.778 ms for the OISST point/region requests. GRIB numerical selections decode the selected field; its encoded view returns the message bytes. Wider Arrow projections increase delivery work. The compressed and uncompressed narrow cases use equal source values; managed layout has similar warm latency in these two cases.

**Table S9: Native-reader session times, including preparation and cleanup (ms). $Q$ counts queries per session. Entries are medians of three independently timed sessions, with fresh results for every request.**

| Format | Reader state | $Q = 1$ | $Q = 10$ | $Q = 100$ |
|---|---|---|---|---|
| GRIB | Reopened | 1541.272 | 15741.510 | 160277.891 |
| GRIB | Retained | 1527.584 | 1579.231 | 2276.949 |
| HDF5 | Reopened | 0.492 | 3.404 | 33.057 |
| HDF5 | Retained | 0.482 | 0.809 | 3.981 |

Table S9 uses the 4,598-message GRIB and the HDF5 source with 160 time slices. Retained native readers reuse handles and locators, while each GRIB query still reads and decodes a fresh message. Preparation is charged within each session. Operating-system and database caches remain enabled in both states. Reuse substantially reduces repeated discovery, showing why reader state accompanies every comparison. Six managed GEOS full-array requests reach the 64 MiB result limit; successful point and region queries retain their measured latencies separately.

*Managed Query Sessions.* The managed sessions in main-paper Section 6.3.2 use the 4,598-message GRIB and 537.68 MB HDF5 sources. Each session includes preparation, fresh-result requests and cleanup. Let $S_r(Q)$ denote session wall time for $Q$ requests in round $r$. The median cost per additional request over three paired rounds is

$$\Delta = \mathrm{median}_r[(S_r(100) - S_r(10))/90].$$

It is computed within each paired round before taking the median; it is not obtained by subtracting the separately reported session medians. These managed-session measurements are distinct from the native-reader sessions in Table S9.

## S5 COLLECTION SCALABILITY

This section provides all sixteen collection configurations and their concurrency protocol for main-paper Section 6.3.3.

**Table S10: Cross-file query latency and IoTDB-FS throughput gains across 16 configurations. Latencies are request medians (ms). $N$ counts files and $C$ candidate paths. $\ell$ is the side length of nonmatching units, while matching units remain $64 \times 64$. $g$ is throughput with four concurrent requests divided by that with one. † marks a throughput loss.**

| | | | | | | Managed latency | | |
|---|---|---|---|---|---|---|---|---|
| Format | $N$ | $C$ | $\ell$ | Scan | Index | One | Four | $g$ |
| GRIB | 8 | 8 | 64 | 34.465 | 4.131 | 13.399 | 30.689 | 1.830 |
| HDF5 | 8 | 8 | 64 | 6.029 | 2.736 | 18.435 | 32.855 | 1.890 |
| GRIB | 128 | 128 | 64 | 420.603 | 2.841 | 7.169 | 18.883 | 1.327 |
| HDF5 | 128 | 128 | 64 | 160.716 | 4.382 | 9.400 | 40.691 | 0.732† |
| GRIB | 512 | 512 | 64 | 1984.135 | 8.083 | 18.177 | 44.906 | 1.648 |
| HDF5 | 512 | 512 | 64 | 397.836 | 6.295 | 24.160 | 39.819 | 2.041 |
| GRIB | 512 | 8 | 64 | 32.670 | 4.301 | 16.623 | 35.329 | 2.570 |
| HDF5 | 512 | 8 | 64 | 6.714 | 2.981 | 6.676 | 41.437 | 0.553† |
| GRIB | 512 | 32 | 64 | 158.482 | 5.046 | 16.981 | 30.862 | 2.649 |
| HDF5 | 512 | 32 | 64 | 20.295 | 1.160 | 8.223 | 34.460 | 0.819† |
| GRIB | 512 | 128 | 64 | 546.087 | 5.638 | 16.355 | 34.429 | 1.974 |
| HDF5 | 512 | 128 | 64 | 103.850 | 3.810 | 24.580 | 38.350 | 2.384 |
| GRIB | 8 | 8 | 384 | 33.514 | 2.386 | 6.913 | 15.426 | 1.624 |
| HDF5 | 8 | 8 | 384 | 5.530 | 1.433 | 6.445 | 17.945 | 1.103 |
| GRIB | 8 | 8 | 768 | 45.493 | 2.547 | 10.346 | 14.912 | 3.195 |
| HDF5 | 8 | 8 | 768 | 6.066 | 2.927 | 22.820 | 41.358 | 1.902 |

This study separates discovery work from useful numerical output. Six attribute equalities return two complete 64×64 Float32 units and their masks, totaling 40,960 bytes. Every execution path receives the same candidate set. Total files $N$, candidate logical paths $C$ and nonmatching payload size vary independently; matching units and the result contract remain fixed. Each file contains four native units and one version; all outputs pass native-reference/source checks.

Table S10 retains all sixteen configurations. The $N = C$ series measures collection growth, the fixed-$N$ series varies discovery scope, and the last two size profiles enlarge only nonmatching units. FS-scan and FS-index share the numerical reader. Managed HDF5 reads bound complete files with native readers. Operating-system and database caches remain enabled, and numerical reads within one query execute sequentially. Native contexts retain eight files/datasets, with 2 MiB metadata caches per file and 256 KiB chunk caches per dataset. Managed execution uses four workers, an 8 MiB payload cache, and native-HDF5 handle reuse in both schedules.

One complete campaign covers the sixteen configurations. Two warmup queries precede ten measured requests per native configuration. Managed measurements use two warmup and ten measured batches, each repeating the same query four times with one or four in flight. Latencies are request medians; throughput divides forty completed requests by the summed measured batch time. Thus the table compares serial native references and two managed admission schedules under their stated retained state.

At 512 candidate files, GRIB scanning spends 1979.120 ms in discovery and 3.532 ms in subsequent reading/delivery; indexing reduces discovery to 4.300 ms with a 3.685 ms reading/delivery summary. These independently aggregated phases locate the main saving in discovery.

At $N = C = 8$ and the smallest size profile, GRIB discovery takes 31.857 ms with FS-scan and 1.155 ms with FS-index under matched numerical readers, illustrating the discovery savings discussed in main-paper Section 6.2.2.

*Discovery Scope Controls.* At fixed $N = 512$, expanding $C$ from eight to 512 gives managed latency growth of 1.09 for GRIB and 3.62 for HDF5, compared with 60.73/59.25 for scanning and 1.88/2.11 for indexing. From nonmatching-unit side length $\ell = 384$ to 768, recorded median ratios are 1.50/3.54 managed, 1.07/2.04 indexed, and 1.36/1.10 scanning. Medians across all three size profiles are nonmonotonic; these ratios describe the selected endpoints rather than a payload-size trend.

*Concurrent Query Costs.* Within this campaign, thirteen cases gain throughput under concurrent admission; all eight GRIB cases improve. Request medians rise in all configurations.

Throughput ratios for four versus one request in flight span 1.33–3.20 for GRIB and 0.55–2.38 for HDF5; the three HDF5 losses are 26.8%, 44.7%, and 18.1%. At $N = C = 512$ and the smallest size profile, mean four-query batch times fall from 86.006/99.099 to 52.179/48.549 ms (GRIB/HDF5), while request medians rise from 18.177/24.160 to 44.906/39.819 ms. Dividing each unrounded mean batch time by four gives amortized wall-time costs of 21.50 to 13.04 ms for GRIB and 24.77 to 12.14 ms for HDF5. This is inverse throughput, not individual response latency or processor time. Native paths provide serial latency references; the concurrency comparison concerns managed scheduling. The ten measured batches per schedule describe variability under the recorded shared-host load, rather than independent replications of the concurrency effect.

## S6 HISTORICAL QUERY COSTS

This section adds discovery and reading/delivery phase costs to the historical-query results in main-paper Section 6.3.4.

Historical selection introduces two independent quantities: retained publications and returned data. The study holds eight candidate logical paths and four native units per source. Identical histories repeat content; changed histories update one unit per publication and enlarge the HDF5 files. Latest and the first historical window examine eight source versions and return two native units; the last-two window examines sixteen and returns four, while All examines $8V$ and returns $2V$.

**Table S11: Discovery and reading costs for historical queries (ms), with 32 retained versions per path. Each $d/q$ entry lists discovery time followed by reading, decoding, and delivery time. $K$ counts returned units. Each phase is summarized separately as the median of three round medians.**

| History | Selection | $K$ | Scan $d/q$ | Index $d/q$ | Managed $d/q$ |
|---|---|---|---|---|---|
| **GRIB** | | | | | |
| Identical | Latest | 2 | 29.013/1.773 | 0.472/2.118 | 4.476/5.713 |
| Identical | All | 64 | 842.392/46.939 | 2.046/47.191 | 18.800/188.184 |
| Changed | Latest | 2 | 27.664/1.649 | 0.527/2.260 | 4.394/6.220 |
| Changed | All | 64 | 852.577/47.296 | 2.026/47.033 | 20.718/189.178 |
| **HDF5** | | | | | |
| Identical | Latest | 2 | 4.887/1.413 | 0.446/0.615 | 1.937/8.490 |
| Identical | All | 64 | 190.077/63.528 | 2.094/49.286 | 2.409/199.515 |
| Changed | Latest | 2 | 3.151/0.685 | 0.522/0.644 | 1.939/8.628 |
| Changed | All | 64 | 148.623/46.035 | 2.078/48.148 | 2.231/198.186 |

Table S11 complements the main paper's complete latency table and growth discussion with discovery and reading/delivery phases at the largest retained history. With Latest, all execution paths return 40,960 bytes. All returns 1,310,720 bytes, increasing both numerical access and delivery. Reading/delivery includes the remaining execution work after discovery; separately summarized phases need not sum to a reported total.

For GRIB, scanning visits eight sources for Latest and 256 for All, repeatedly extracting scientific attributes. Its discovery phase grows to 842.392/852.577 ms for identical/changed All histories. Indexing reuses the scientific catalog; its remaining reading/delivery cost increases with output. IoTDB-FS likewise spends most HDF5 All-query time after discovery, in the phase that groups all remaining access and delivery work. The fixed-output selectors and All therefore support different growth interpretations.

The original indexed and managed measurements use three rounds, each with two warmups and ten sequential measured queries per case, after histories are prepared. GRIB reads selected units individually, whereas HDF5 batches them. The original managed HDF5 path uses one native read at a time without the additional native handle cache; FS-index retains its bounded native contexts. Their summaries are medians of the three round medians. Historical FS-scan was collected later on the same inputs with the same helper, clock and resource limits; scan and an independent index control alternate within each case. The repeated/original index latency ratio has a cross-case median of 1.008 and range 0.405–2.562. This records collection variability while preserving the original observations; no measurement is rescaled by that ratio.

These phase results describe the measured retained-cache histories. Stable fixed-output growth concerns a bounded selected source set, while All exposes growing output. Neither growth coefficient alone specifies absolute query cost, which remains explicit in the main latency tables.

*Historical Cost Growth.* Across twelve fixed-output cases, the median ratio of latency at $V = 32$ to that at $V = 8$ is 0.972 for IoTDB-FS, 1.041 for FS-index, and 1.068 for FS-scan. All increases output from 16 to 64 units; the corresponding growth ranges are

3.623–4.682, 3.113–3.847, and 4.014–5.669. These ratios use reported cell medians and describe the recorded campaigns. In the changed HDF5 All case at $V = 32$, managed and scan latencies are 200.500 and 192.133 ms, with round-median standard deviations of 7.393 and 1.872 ms. The managed warm-state pool's sampled accounting peak is 44.77 MiB out of 64 MiB; this is retained-state accounting rather than process memory.

The smaller managed growth ratio also reflects a higher starting latency. Scan and managed observations come from separate collections; their small differences do not isolate path effects, and round-level deviations do not measure variation between collections.

## S7 QUERY LIFECYCLE COSTS

This section supplements main-paper Section 6.3.5 with warm-query protocols and publication/update traces, each under its own timing boundary.

*Warm-Query Sessions.* Filesystem paths share immutable local files, while IoTDB-FS reopens stored versions without importing their payloads again. The initial campaign has 135 sessions and 4,995 measured queries over $Q = 1, 10, 100$. The paired HDF5 follow-up compares scanning, reopened index handles, and retained index handles over 81 sessions and 2,997 measured queries. Managed observations remain from the initial campaign. For each source/path, a first request and two warmups precede the measured sessions; path and session-length order rotate across three rounds. The reported mean uses each round's session wall time divided by $Q$.

FS-scan and GRIB FS-index reopen native handles per query. HDF5 FS-index retains up to eight file/dataset handles with a 1 MiB chunk cache per dataset; every request checks its source binding and performs a native value read, which the retained chunk cache may satisfy. Initial opening and final closing of retained HDF5 handles are recorded separately. IoTDB-FS HDF5 uses `iotdb-indexed` with retained caches. The managed $Q = 100$ sessions record zero database payload bytes; database heads and owner journals remain unchanged. Warm serving uses stored sources and prepared indexes, with source verification, recovery, and auditing outside its timing boundary. Recovery takes 213–222 s, and large GRIB's first managed request averages 104.559 ms after recovery with caches retained. For restart workloads, recovery contributes an additional measured phase.

*Historical Import Costs.* Table S12 reports the historical two-path lifecycle from importing local files into each backend through querying. Ninety lifecycles cover five sources, two paths and three query counts, with 3,330 checked results. Each continuous clock spans copy entry through final result consumption, including file synchronization and close; parent-directory synchronization follows it and startup precedes it. Retained content permits deduplication across publications.

These historical totals show GRIB discovery savings accumulating with repeated queries, whereas managed HDF5 preparation remains prominent because native array access is inexpensive.

*Dynamic Index Maintenance.* The incremental study uses eight initial files and sixteen updates, each followed by $q$ queries ($Q = 16q$ per trace). Every result contains two numerical units and masks

**Table S12: Total time to import files and run point queries (s), including placement in the target backend. Entries are medians of three complete runs. $Q$ counts queries after import, and $t$ counts HDF5 time slices.**

| Source | Path | $Q = 1$ | $Q = 10$ | $Q = 100$ |
|---|---|---|---|---|
| GRIB, 22 messages | Native FS | 0.045 | 0.238 | 2.184 |
| GRIB, 22 messages | IoTDB-FS | 2.830 | 2.719 | 2.960 |
| GRIB, 4,598 messages | Native FS | 2.775 | 15.684 | 147.564 |
| GRIB, 4,598 messages | IoTDB-FS | 43.264 | 41.262 | 43.649 |
| HDF5, $t = 2$ | Native FS | 0.031 | 0.031 | 0.058 |
| HDF5, $t = 2$ | IoTDB-FS | 2.612 | 2.701 | 2.863 |
| HDF5, $t = 160$ | Native FS | 1.346 | 1.357 | 1.388 |
| HDF5, $t = 160$ | IoTDB-FS | 43.428 | 45.327 | 43.127 |
| HDF5, $t = 1024$ | Native FS | 7.094 | 6.910 | 7.489 |
| HDF5, $t = 1024$ | IoTDB-FS | 285.190 | 288.622 | 290.711 |

(40,960 bytes). Larger query blocks include 10% first-publication selections. Corresponding paths consume identical input bytes. Its clock spans initialization, external-file acquisition, publication, maintenance, queries, and finalization. Three repetitions across both formats and all three paths yield 54 traces and 31,968 checked queries.

**Table S13: Total time for file updates and queries (s). Each run includes initialization, initial file publication, 16 updates with $q$ queries after each update, and finalization. Entries are arithmetic means of three complete runs.**

| Format | $q$ | FS-scan | FS-index | IoTDB-FS |
|---|---|---|---|---|
| GRIB | 1 | 0.967 | 0.750 | 45.170 |
| GRIB | 10 | 4.134 | 0.888 | 44.890 |
| GRIB | 100 | 36.515 | 3.107 | 51.321 |
| HDF5 | 1 | 0.659 | 0.654 | 54.264 |
| HDF5 | 10 | 1.081 | 0.729 | 55.097 |
| HDF5 | 100 | 5.499 | 1.696 | 63.275 |

Each indexed update processes only the new file. The managed write path adds 24 empty publications alongside 24 intended content publications per trace; these events contribute to its complete cost under the matched query-visible content contract. At one query per update, HDF5 scanning and indexing are nearly tied. Indexed discovery limits growth in trace cost as query count increases, while managed publication remains the dominant term. Shared database background activity accompanies these descriptive measurements.

At $q = 100$, the 51.321 s GRIB managed trace comprises initialization (1.625 s), initial publication (14.882 s), updates (27.909 s), queries (6.748 s), finalization (0.146 s), and 0.011 s outside these phases. Publication dominates this import workload, including the empty commits. The total comes from a continuous wall-clock measurement.

Each file contains four 64×64 native units. Across 144 indexed updates per format, extraction/validation averages 5.009/1.927 ms for GRIB/HDF5. The complete publication/update operation averages 26.809/24.109 ms and also includes file copying, synchronization,

and catalog commit. Final catalogs occupy 124 KiB with unit indexing or 100 KiB for source/version records. Latest and first-version queries pass the native reference checks.

## S8 REMOTE QUERY PERFORMANCE

This section supplements the simulated distributed-query study in main-paper Section 6.3.6.

Query and storage containers share one host and network namespace, with separate two-core budgets and 2/24 GiB memory limits, respectively. Storage hosts one DataNode, one ConfigNode, and an NFS server. Loopback traffic has 0/1/10 ms added RTT and a 1 Gbit/s limit per direction, simulating delay without physical cross-host deployment. The 128-file, single-version corpus uses the smallest size profile. Four queries with six attribute equalities return two numerical units and masks (40,960 bytes), executed sequentially or with four requests in flight. Each setting has one initial batch, two warmups, and ten measured four-request batches. Throughput divides forty completions by their summed batch makespans. Read-only NFSv4.1 uses `actimeo=0` and `lookupcache=none`, disabling attribute/name caches while retaining data, library, and database caches. IoTDB-FS uses database access for GRIB and managed discovery with native NFS reads for HDF5. Both schedules retain the collection-experiment caches and four managed workers/readers.

**Table S14: Warm-query latency medians and throughput under simulated round-trip delay (RTT). $c$ counts concurrent requests. Latency is rounded to three decimals and throughput to four. A dash marks the configuration that timed out before warm measurements.**

| | | Latency (ms) | | Throughput (queries/s) | |
|---|---|---|---|---|---|
| RTT (ms) | Path | $c = 1$ | $c = 4$ | $c = 1$ | $c = 4$ |
| **GRIB** | | | | | |
| 0 | FS-scan | 1883.187 | 2311.234 | 0.5542 | 1.7247 |
| 0 | FS-index | 13.963 | 34.956 | 67.8305 | 108.7817 |
| 0 | IoTDB-FS | 4.916 | 12.399 | 182.0287 | 277.0863 |
| 1 | FS-scan | 12081.090 | 11100.158 | 0.0836 | 0.3592 |
| 1 | FS-index | 135.168 | 124.232 | 7.4159 | 32.2790 |
| 1 | IoTDB-FS | 4.697 | 12.638 | 197.6437 | 279.9555 |
| 10 | FS-scan | 86337.009 | 94017.040 | 0.0115 | 0.0416 |
| 10 | FS-index | 989.566 | 981.578 | 1.0123 | 4.0741 |
| 10 | IoTDB-FS | 4.547 | 12.751 | 198.6381 | 284.0667 |
| **HDF5** | | | | | |
| 0 | FS-scan | 2870.325 | 8073.558 | 0.3381 | 0.4823 |
| 0 | FS-index | 13.731 | 22.986 | 71.8154 | 168.9181 |
| 0 | IoTDB-FS | 11.260 | 32.951 | 81.7453 | 78.2338 |
| 1 | FS-scan | 32682.343 | 84465.488 | 0.0306 | 0.0485 |
| 1 | FS-index | 103.664 | 107.024 | 9.6221 | 37.1835 |
| 1 | IoTDB-FS | 69.041 | 174.105 | 14.2833 | 14.8282 |
| 10 | FS-scan | 240141.027 | — | 0.0042 | — |
| 10 | FS-index | 965.984 | 965.516 | 1.0367 | 4.1364 |
| 10 | IoTDB-FS | 599.597 | 1485.608 | 1.6718 | 1.6842 |

*Format-Dependent Reuse.* Table S14 lists all configurations. At 10 ms, managed GRIB's first sequential request takes 193.698 ms, versus a 4.547 ms warm median. HDF5 takes 4,303.661/599.597 ms, respectively, with numerical reads still crossing NFS. Initial batches retain OS/database caches, and later requests can reuse earlier state, so these are not cold-disk measurements.

For GRIB at 10 ms with one request in flight, discovery takes 483.224 ms for FS-index and 1.276 ms for IoTDB-FS. Reading and delivery take 507.291 ms and 3.250 ms, respectively. These phase medians include source checks and NFS costs.

Managed HDF5 discovery at 10 ms takes 1.326/2.933 ms with one/four requests in flight. Reading and delivery take 598.232 ms and 1,482.689 ms, respectively. The four-request p95 is 2,379.095 ms. The native reader holds a process-wide lock, consistent with limited throughput gains as latency rises. The campaign does not isolate the lock's contribution, and FS-index achieves higher concurrent throughput.

*Storage-Side Traffic.* Table S15 sums archived DB/NFS request and reply bytes at 10 ms. The envelopes include out-of-clock validation/profile work and shared background traffic, so totals measure neither file-payload/disk bytes nor bandwidth utilization. Archived `warm_wire` records retain directional byte, packet, and drop counts for all completed settings.

**Table S15: Storage-side network traffic at 10 ms round-trip delay (MiB). Totals cover 40 warm queries, validation, and background traffic. $c$ counts concurrent requests. A dash denotes no warm result.**

| Format | $c$ | FS-scan | FS-index | IoTDB-FS |
|---|---|---|---|---|
| GRIB | 1 | 187.192 | 2.082 | 0 |
| GRIB | 4 | 191.419 | 2.130 | 0 |
| HDF5 | 1 | 833.693 | 2.039 | 1.246 |
| HDF5 | 4 | — | 2.085 | 1.247 |

HDF5 scanning at 10 ms with four requests in flight reaches the 300-second initial-request limit and has no warm result. All 1,820 completed requests (1,400 warm) pass native-reference checks. The 0/1 ms and 10 ms runs form one descriptive campaign without independent repetitions, with containers restarted under unchanged settings. Initialization and source publication precede query timing.

## S9 DATASET INVENTORY

Table S16 describes the datasets accompanying main-paper Section 6.1 and Table 3 using a per-file size convention. It connects the semantic checks and performance studies to their actual source sizes, layouts, and query roles.

**Table S16: Dataset characteristics corresponding to main-paper Table 3. Sizes describe individual encoded files in decimal MB, excluding indexes and source documentation. Ranges cover the evaluated file-size variants. Public inputs retain their native or documented converted layouts. Generated inputs provide separate scale and publication controls.**

| Dataset and source | Data domain | File size (MB) | Tested layout and extent | Query role |
|---|---|---|---|---|
| GFS [source] | Weather forecasts | 2.447–549.607 | GRIB2 file with 22–4,598 messages; 0.25° fields | Mapping, point/region extraction, discovery and reuse |
| OISST v2.1 [source] | Observation-based sea-surface temperature analysis | 1.529–1.534 | Daily netCDF-4/HDF5 file with a global sea-surface temperature grid | Mapping, time/spatial selection, native granularity |
| GEOS-5.7.2 [source] | Atmospheric analysis | 445.984–448.792 | netCDF-4/HDF5 file with a temperature array of 1×42×721×1152 values | Coordinate mapping, point/region chunk selection |
| Beijing [source] | Air quality and weather observations | 4.401–4.648 | Station CSV → flat Arrow; 35,064 hourly rows per file | Agent-assisted mapping, station/time windows |
| Jena [source], [Arrow] | Weather observations | 4.428 | Single nested Arrow file; 21 channels × 52,704 grid positions | Agent-assisted mapping, irregular timestamps, ordinal windows |
| Appliances [source] | Household energy and environment | 4.587 | CSV → a flat Arrow file; 19,735 rows and 29 columns | Row/time selection, projections and residual counts |
| LOOP Seattle [Arrow] | Transport benchmark | 11.327 | Single nested Arrow file; 323 series × 8,760 hourly steps | Series/time windows and known-position controls |
| Solar [Arrow] | Energy benchmark | 28.808 | Single nested Arrow file; 137 series × 52,560 ten-minute steps | Series/time windows and known-position controls |
| KDD Cup 2018 [Arrow] | Air-quality benchmark | 11.777 | Single nested Arrow file; 270 series, 9,504–10,920 hourly steps, missing values | Series/time windows and residual counts |
| Saugeen [source], [Arrow] | River flow | 0.096 | Single nested Arrow file; 23,741 daily values | Source alignment, direct-slice control |
| Temperature–Rain [source], [Arrow] | Temperature observations and rainfall forecasts | 93.877 | Single nested Arrow file; 32,072 series × 725 positions | Time-label alignment, identifier-to-row selection |
| Electricity [source] | Electricity consumption | 207.613 | Tabular → a nested Arrow file, 370 channels; compressed variant: 93.676 MB | Single-file windows, wider projections, compression control |
| Generated HDF5 arrays | Array-size profiles | 6.729–3,441.125 | HDF5 file with $t \times 1 \times 728 \times 1152$ values; $t = 2, 160, 1024$ | Point/region queries, sessions and state reuse |
| Generated GRIB/HDF5 collections | Archive and publication profiles | — | GRIB/HDF5 file with immutable versions; matching units 64×64, nonmatching side lengths 64/384/768 | Collection, concurrency, history, incremental maintenance and remote access |

Source links identify the public data source; Arrow links identify the tested GIFT-Eval representation. GIFT-Eval inputs use the archived publisher revision: [Arrow]. Domain labels identify source context; the verified mappings remain those in main-paper Table 3. The collection row spans independently generated profiles and has no single representative file size. Collection and history settings are separate sweeps, not a full $N \times V$ grid.